Technical Report: Periodic-1D Higher-Rank Composite-Band Alignment and Locality

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Schema-free development calculation

1 Scope and evidence boundary

This report covers the finite one-dimensional higher-rank controlled calculation used by the Paper C development program. It is calculation-owned and publication-independent. The current implementation and ignored summaries are replaceable development work, not a retained run, convergence result, or evidence package.

2 Finite model inventory

The selected ladder uses retained ranks $r \in \{3, 4, 8, 16\}$, rotation depths $L \in \{1, 2, 4, 8\}$, ambient dimension $2r$, construction meshes 513, 1025, and 2049, an independent evaluation mesh of size 8191, and hopping supports $0, 1, 2, 4, 8, 16, 32, 64, 128$.

Four alignment controls remain separate: no alignment, one global unitary, pointwise complete unitaries, and the known generating inverse. The known inverse is an oracle control and is not evidence of practical recoverability.

3 Analytic qualification

For the selected fixed-subspace real-rotation parent, the retained–exterior separation has the exact value

$$\Delta_{\mathrm{iso}} = 2.10 E_{\mathrm{ref}}. \tag{1}$$

The calculation-owned finite-grid utility reproduces this value on a grid containing the attaining momentum. Sampling is a sanity check; the protocol derivation is the analytic certificate.

4 Development computation

The schema-free producer evaluates all sixteen rank/depth cases, the four alignment controls, transport and sewing diagnostics, Fourier support, interpolation and truncation errors, refinement records, and sampled support–error Pareto classifications. It writes a compact replaceable summary rather than a retained package.

5 Verification boundary

Software tests cover model construction, alignment selection and failure dispositions, transport, scalar ledgers, numerical exception handling, and one complete schema-free case. The analytic-control checker is not a producer-independent reconstruction of the complete result. Strict serializers, schemas, source binding, and a full independent verifier remain required.

6 Limitations and nonclaims

No two- or three-dimensional higher-rank calculation exists. Moving retained subspaces, complex unitary transformations, and restricted-circuit fitting remain prospective. The controlled family is topologically trivial and is not a material model. Current work establishes neither continuum convergence nor scientific validation.

7 Next gate

The next gate is a strict versioned input/result contract and a producer-independent direct-formula verifier. Only after those pass may a separately authorized retained execution be considered. Analysis and plots must consume verified output rather than rerun the producer implicitly.